

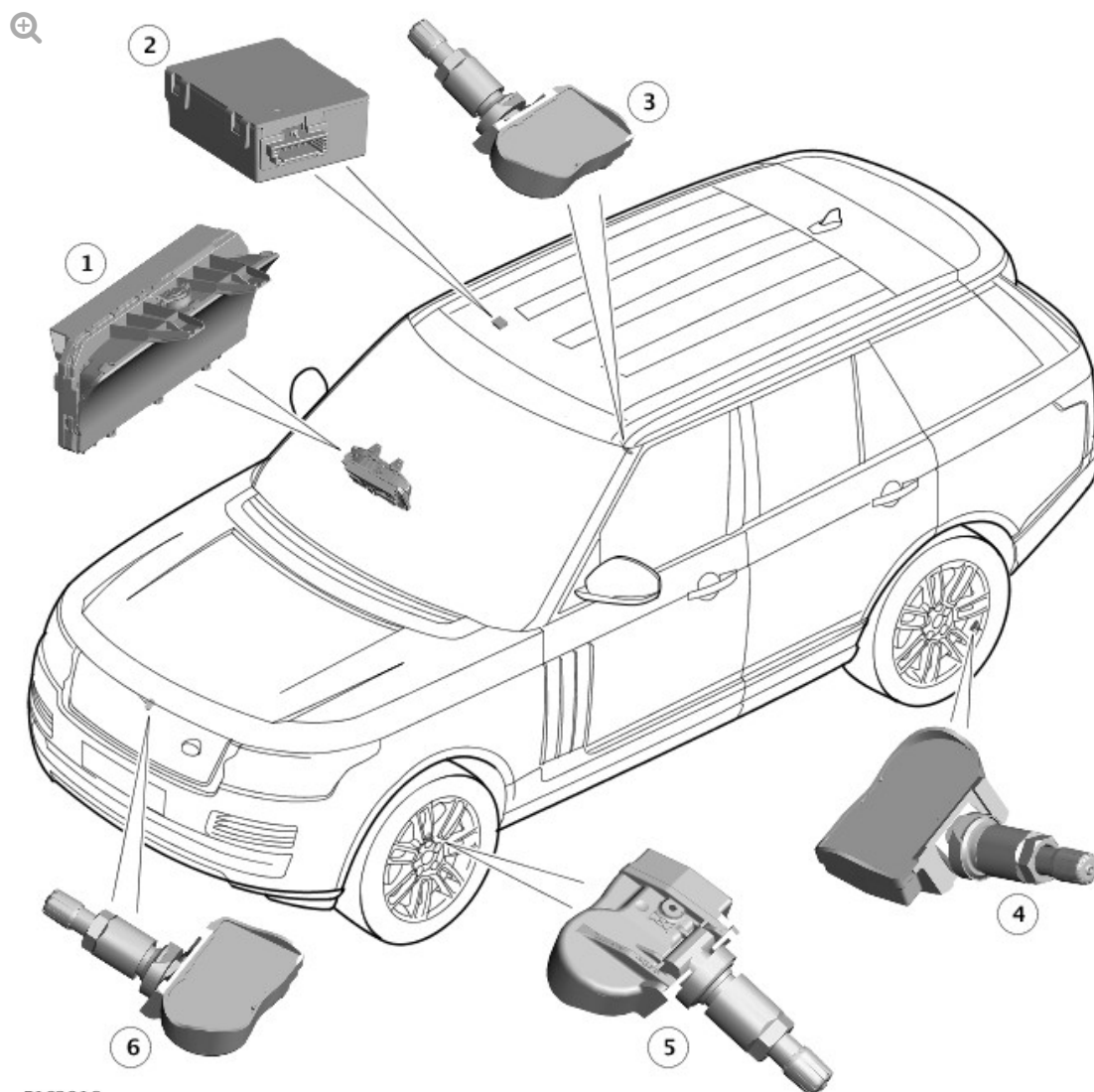
2016.0 RANGE ROVER (LG), 204-04

# WHEELS AND TIRES

DESCRIPTION AND OPERATION

## COMPONENT LOCATION

### TIRE PRESSURE MONITORING SYSTEM (TPMS)



E182916

| ITEM | DESCRIPTION                                             |
|------|---------------------------------------------------------|
| 1    | Instrument Cluster (IC)                                 |
| 2    | Tire Pressure Monitoring System Control Module (TPMSCM) |
| 3    | Tire pressure sensor - right rear                       |
| 4    | Tire pressure sensor - left rear                        |
| 5    | Tire pressure sensor - left front                       |
| 6    | Tire pressure sensor - right front                      |

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## OVERVIEW

### TIRE PRESSURE MONITORING SYSTEM (TPMS)

The purpose of the Tire Pressure Monitoring System (TPMS) is to help the driver to keep the vehicle's tire pressures at the optimum level to:

- Improve fuel consumption.
- Maintain ride and handling characteristics.
- Reduce the risk of rapid tire deflation, which may be caused by under-inflated tires.
- Comply with legislation in relevant markets.

The TPMS measures the pressure in each of the tires on the vehicle and issues warnings to the driver if any of the pressures deviate from defined tolerances.

**NOTE:**

During a tire 'blow-out' a very rapid reduction in pressure happens. The system is not intended to warn the driver of a 'blow out', as it is not possible to give the driver enough warning that such an event is happening, due to its short duration. The TPMS helps the driver to keep the tires at the correct pressure and this tends to reduce the likelihood of a tire 'blow out' happening.

TPMS is inhibited when the vehicle is in Transit mode. For more details on Transit mode, refer to the Pre-Delivery Inspection (PDI) manual.

A single TPMS hardware configuration is used. TPMS status information is relayed to the driver by a message which appears in the Instrument Cluster (IC) message center together with an amber warning indicator.

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**DESCRIPTION****TIRE LOCATION****CAUTION:**

Take care when you remove and refit tires to ensure that the tire pressure sensor is not damaged.

Because of the requirement for different pressure targets and thresholds for the front and rear tires, the Tire Pressure Monitoring System (TPMS) can identify the location of the sensors on the vehicle, and assign received tire pressure sensor identification to a specific position on the vehicle, that is to say, front left, front right, rear left or rear right.

The TPMS does the tire sensor location automatically using an auto-location function. This function does not need any manual intervention by the driver.

The TPMS can automatically learn the position of tire sensors on the vehicle if the tire pressure sensors or their positions are changed on the vehicle.

The tire-learn and location process is ready to start when the vehicle has been stationary or is traveling at less than 20 km/h (12.5 mph) for 15 minutes. This is known as Parking mode. The learn/locate process needs the vehicle to be driven at speeds of more than 20 km/h (12.5 mph) for 10 minutes. If the vehicle speed reduces to below 20 km/h (12.5 mph), the learn process timer is suspended until the vehicle speed increases to more than 20 km/h (12.5 mph), after which time the timer is resumed. If the vehicle speed stays below 20 km/h (12.5 mph) for more than 15 minutes, the timer is set to zero and the process starts again.

The TPMS can automatically detect, under all operating conditions, the following:

- One or more tire pressure sensors have been replaced.
- One or more tire pressure sensor identifications are missing.
- One or more tire pressure sensors have stopped transmitting.
- Identifications from tire pressure sensors which do not belong to the vehicle.
- One or more of the wheels in use on the vehicle has changed position.
- One or more incompatible tire pressure sensors have been fitted.
- A full-sized spare wheel is swapped with a running wheel.

**NOTE:**

The spare wheel sensor ID must be written into the Tire Pressure Monitoring System Control Module (TPMSCM) by diagnostics and cannot be learned automatically.

If the tire pressure sensors fitted to the running wheels are changed, the

TPMS can learn the new sensor identifications automatically. The learn function requires no manual intervention by the driver.

**NOTE:**

You must service the TPMS valve using the appropriate service kit each time you dismount the tire, to ensure an airtight seal. Follow the instructions about using this kit.

## TIRE PRESSURE MONITORING SYSTEM CONTROL MODULE (TPMSCM)

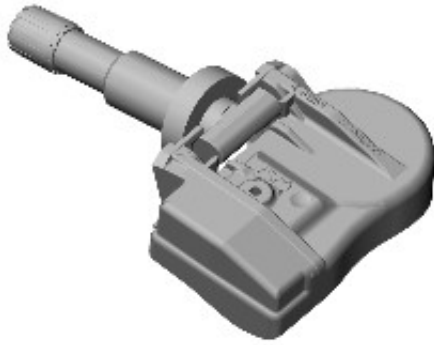


E164435

The Radio Frequency (RF) receiver is integrated into the Tire Pressure Monitoring System Control Module (TPMSCM), which is mounted on the headliner next to the rear overhead console and connected to the vehicle harness using a six-pin plug.

The TPMSCM receives transmissions from each of the tire pressure sensors which are sent by means of an internal antenna in each of the sensors. This information is then decoded and analyzed and any required warnings along with current tire pressure information is communicated on the High Speed (HS) Controller Area Network (CAN) powertrain bus.

## TIRE PRESSURE SENSOR



E164794

The Tire Pressure Monitoring System (TPMS) system uses 'active' tire pressure sensors which are mounted on each wheel rim, inside the tire cavity. The sensor is retained in position by the valve attachment to the wheel structure. The sensors transmit their Radio Frequency (RF) signals at 433 MHz.

The sensors periodically measure the pressure and temperature of the air inside the tire plus the centripetal acceleration acting on the sensor. These measurements are transmitted periodically to the RF receiver located behind the rear overhead console.

The tire pressure sensors are self-contained units as they do not have any electrical connections going into or coming out of the sensor.

#### **CAUTION:**

You must follow the care points detailed in the 'Tires' section of this chapter to avoid damaging the sensor.

If you replace the sensor, you must also replace the nut, seal and washer and tighten the sensor to the correct torque value as given in the Service Repair manual.

For additional information, refer to: [Tire Low Pressure Sensor](#) (204-04 Wheels and Tires, Removal and Installation).

The RF transmission from the sensor contains a unique identification code in its transmission data, so that the TPMS can identify the tire sensor on the vehicle. If you replace a sensor on a 'running' wheel, you need to program the identification for that sensor into the Tire Pressure Monitoring System Control Module (TPMSCM), using a Land Rover approved diagnostic system, otherwise the Tire Pressure Monitoring System (TPMS) will not monitor the wheel that is fitted with the sensor you replaced. The code is provided on a label with the complete wheel and tire assembly when new and is also printed on the casing of each sensor.

To conserve battery power, the tire sensor uses different transmission rates when the wheel is stationary or moving. The wheel speed required to change between the stationary and moving transmission rates is 20 km/h (12.5 mph) to allow for slow driving.

#### INSTRUMENT CLUSTER (IC) INDICATIONS



You can display the recommended and the actual tire pressures in the IC message center as well as changing the pressure units (KPa, PSI or bar) and you can select light or heavy load in the Vehicle Info and Tire Pressure Check Instrument panel menus.

The warning indications to the driver are common on all vehicles fitted with

TPMS. Warnings are conveyed by an amber warning indicator together with a text message, a supporting graphic, or both of these which appear in the message center.

The warning indicator and message center are driven by Controller Area Network (CAN) messages from the Tire Pressure Monitoring Control Module (TPMCM). The warning indicator is illuminated by the cluster software continuously when the tire pressure is low and will flash for 75 seconds in the case of a sensor failure or a system fault.

CONTROLLER AREA NETWORK (CAN)

The Tire Pressure Monitoring Control Module (TPMCM) sends and receives a number of digital messages through the CAN bus. The received messages are used to operate the Tire Pressure Monitoring System (TPMS). The transmitted messages comprise of TPMS status, warning and pressure information and requests to the IC to turn on or off warning indicators and/or display messages in the message center.

The TPM module transmits the messages shown in the following table:

Recommended tire pressure data including heavy load and/or high speed valuesIC

| MESSAGE                                                                                 | RECEIVED BY                           |
|-----------------------------------------------------------------------------------------|---------------------------------------|
| TPMS diagnostic response                                                                | Land Rover approved diagnostic system |
| TPMS amber warning indicator request at 25% (or 20% depending on market) tire deflation | Instrument Cluster (IC)               |
| TPMS message display request                                                            | IC                                    |
| Current tire pressures                                                                  | IC                                    |

DIAGNOSTICS

The Tire Pressure Monitoring Control Module (TPMCM) has a diagnostic connection through the Controller Area Network (CAN) Bus which allows a Land Rover approved diagnostic system to retrieve the system status and



faults.

Additionally, an on-board diagnostic routine within the TPMCM constantly monitors the system and alerts the driver to system faults by turning on the amber warning indicator and/or displaying a message in the IC message center.

## FAULT DETECTION

If a sensor fails, the amber warning indicator in the Instrument Cluster (IC) flashes and then stays on and a message 'XX Tire Pressure Not Monitored' appears in the IC message center.

### NOTE:

'XX' means the wheel position on the vehicle as follows:

- Front left
- Front right
- Rear left
- Rear right

If more than one sensor fails or the TPMS develops a system fault, the amber warning indicator flashes for 75 seconds and then stays on and a message 'Tire Pressure Monitoring System Fault' appears in the message center. This fault could also happen if RF interference near the vehicle affects the signal reception of the system. When the interference stops, the fault is automatically cancelled and the TPMS operates normally.

If a tire pressure sensor battery fails, the sensor stops transmitting and, after the missing sensor timer expires, the TPMS transmits a message to display 'FL Tire Pressure Not Monitored' for example in the message center. You must interrogate the TPMS using a Land Rover approved diagnostic system to determine the cause of the message. If the sensor has failed, you must replace it and remove the stored fault flags using a Jaguar-approved

diagnostic system.

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## OPERATION

Each time the vehicle is driven, the TPMS learns the locations of the following four wheel-mounted sensors by combining Radio Frequency (RF) transmissions with data from the Anti-lock Brake System (ABS) control module:

- Front left
- Front right
- Rear left
- Rear right

### NOTE:

An ABS fault does not directly cause the TPMS to stop operating. However, if an ABS fault persists, then a TPMS warning may also appear in the IC.

The system enters parking mode after the vehicle speed has been less than 20 km/h (12.5 mph) for 15 minutes. In Parking mode, the tire pressure sensors transmit a coded signal to the TPM module once every 13 hours. If the tire pressure decreases by more than 7 kPa (1 PSI; 0.07 bar), the sensor transmits more often to show that the tire pressure is reducing.

The process of locating the sensors is known as 'Auto Location' and takes up to 10 minutes to complete. During this period, the tire sensors transmit at regular intervals, that is, once every 16 seconds, while the vehicle speed is above 20 km/h (12.5 mph). For the remainder of the drive cycle, the tire sensors transmit once every 64 seconds while the vehicle speed is above 20 km/h (12.5 mph) or, if a change in tire pressure is sensed, until the vehicle stops or the vehicle speed is less than 20 km/h (12.5 mph) for 15 minutes

and the system returns to parking mode.

At 25% deflation (or 20% for the EU market), the amber warning indicator in the IC comes on and an appropriate message and/or graphic appears in the Instrument Cluster (IC) message center.

## TIRES

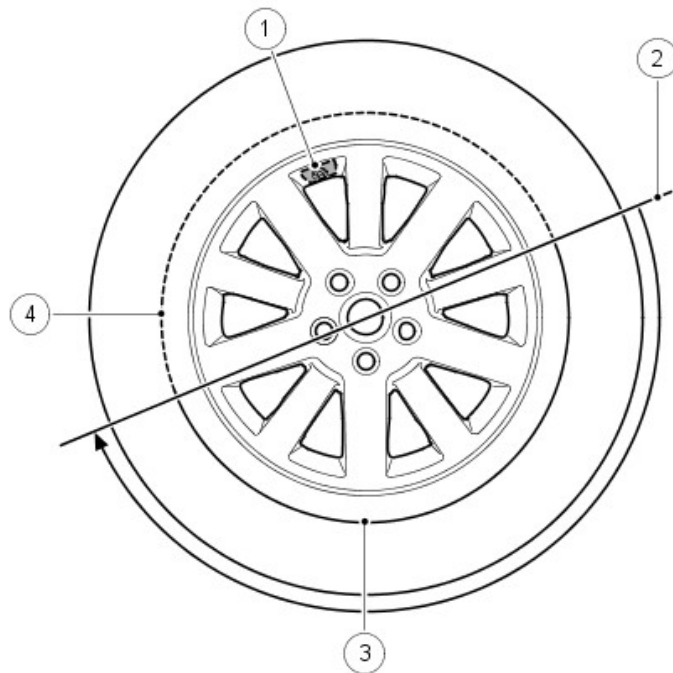
### CAUTION:

Take care when you remove and refit tires to ensure that the tire pressure sensor is not damaged.

### NOTE:

You must service the TPMS valve using the appropriate service kit, each time you dismount the tire, to ensure an air tight seal. Follow the instructions about using this kit.

If the metal valve stem is damaged, you can replace it independently of the tire pressure sensor as it is a separate component so you do not need to replace the entire sensor assembly.



E45549

| ITEM | DESCRIPTION                                      |
|------|--------------------------------------------------|
| 1    | Tire valve and pressure sensor                   |
| 2    | Tire fitting/removal tool initial start position |
| 3    | High tire and bead tension area                  |
| 4    | Low tire and bead tension area                   |

When you remove the tire, you must not use the bead breaker within 90 degrees of the tire valve in each direction.

When you use the tire removal machine, you must position the fitting arm at the start position as shown in the tire changing illustration. You can then rotate the wheel through 180 degrees in a counterclockwise direction. This relieves the high tension from the tire bead and allows you to manually pull the remaining 180 degrees of the tire from the rim.

When you refit the tire, position the fitting arm as shown. Rotate the tire and take care that the bead on the low tension side of the tire does not damage the sensor.

TREAD ACT - NORTH AMERICAN SPECIFICATION (NAS) ONLY

Vehicles supplied to the North American markets must comply with the legislation of the Transport Recall Enhancement, Accountability and Documentation (TREAD) act. Part of the requirement of the TREAD act is for the vehicle to display a label, positioned on the driver's side B-pillar, which defines the recommended tire inflation pressure, seating capacity, and the maximum combined weight of the occupants and cargo the vehicle can safely carry. This label is specific to an individual vehicle and it is installed on the production line.

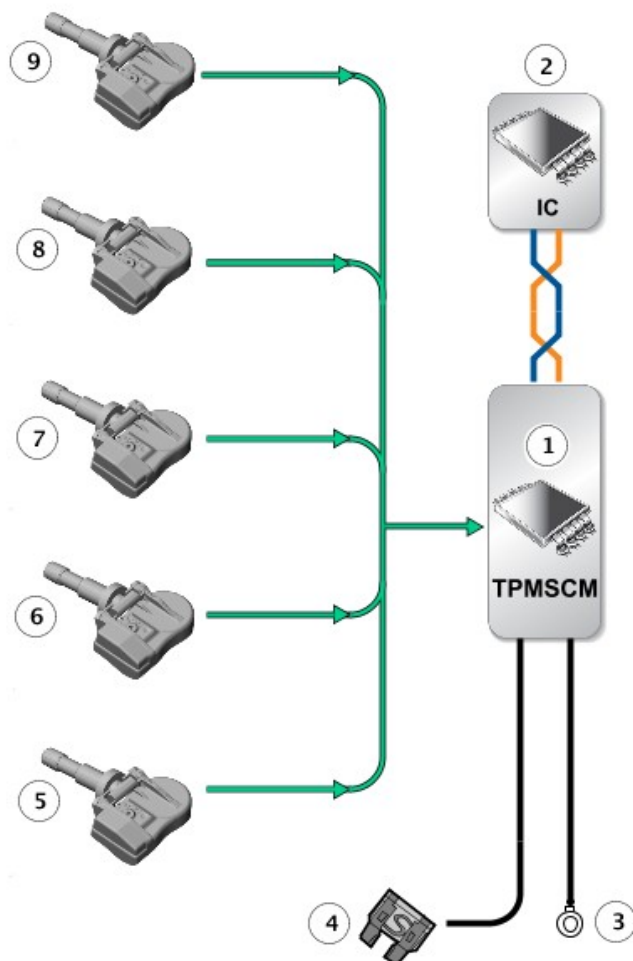
This label must not be removed from the vehicle. The label information only defines the specification of the vehicle as it came off the production line. It does not include dealer or owner fitted accessory wheels and tires of differing size from the original fitment.

#### **NOTES:**

- If tires and wheels of a non-standard size are fitted to the vehicle, the car configuration file must be updated using a Land Rover approved diagnostic system.
- If the label is damaged or removed for body repair, it must be replaced with a new label specific to that vehicle. A new label is requested from Land Rover parts and will be printed specifically for the supplied VIN of the vehicle.

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## **CONTROL DIAGRAM**



E174869



A = HARDWIRED; AN = HIGH SPEED (HS) CONTROLLER AREA NETWORK (CAN) POWERTRAIN BUS; F = RADIO FREQUENCY (RF) TRANSMISSION.

| ITEM | DESCRIPTION                                             |
|------|---------------------------------------------------------|
| 1    | Tire Pressure Monitoring System Control Module (TPMSCM) |
| 2    | Instrument Cluster (IC)                                 |
| 3    | Ground                                                  |
| 4    | Power supply                                            |
| 5    | Tire Pressure Sensor – left front                       |
| 6    | Tire Pressure Sensor – right front                      |
| 7    | Tire Pressure Sensor – left rear                        |
|      |                                                         |

|   |                                                                     |
|---|---------------------------------------------------------------------|
| 8 | Tire Pressure Sensor – right rear                                   |
| 9 | Tire Pressure Sensor – spare (if a full-size spare wheel is fitted) |